University of Ottawa

Department of Mathematics and Statistics

MAT 1302D: Mathematical Methods II
Professor: Hadi Salmasian

First Midterm Test
February 4, 2010

Version B

Surname _____ First Name _____

Student # _____ DGD (1–4) _____

Instructions:

(a) You have 80 minutes to complete this exam.
(b) The number of points available for each question is indicated in square brackets.
(c) Unless otherwise indicated, you must justify your answers to receive full marks.
(d) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will **not** be considered for marks.
(e) Write your student number at the top of **each page** in the space provided.
(f) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(g) You should write in **pen**, not pencil
(h) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	5	5	3	4	4	4	25
Grade							

Student # _____ MAT 1302D First Midterm Test

1. **[5 pts]** Find the general solution to the matrix equation

$$\begin{bmatrix} 2 & -1 & 2 & 0 \\ 6 & -3 & 3 & 1 \end{bmatrix} x = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

and write your answer in **vector parametric form**.

Since A is 2x4, we have $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$, i.e. a 4x1 vector.

The matrix equation $Ax = b$ corresponds to the linear system

$$\begin{cases} 2x_1 - x_2 + 2x_3 = 1 \\ 6x_1 - 3x_2 + 3x_3 + x_4 = -3 \end{cases}$$

· We row reduce its augmented matrix :

$$\left[\begin{array}{cccc|c} 2 & -1 & 2 & 0 & 1 \\ 6 & -3 & 3 & 1 & -3 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cccc|c} 2 & -1 & 2 & 0 & 1 \\ 0 & 0 & -3 & 1 & -6 \end{array}\right]$$

$$\xrightarrow{R_2 \to -\frac{1}{3}R_2} \left[\begin{array}{cccc|c} 2 & -1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -\frac{1}{3} & 2 \end{array}\right] \xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{cccc|c} 2 & -1 & 0 & \frac{2}{3} & -3 \\ 0 & 0 & 1 & -\frac{1}{3} & 2 \end{array}\right]$$

$$\xrightarrow{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{cccc|c} 1 & -\frac{1}{2} & 0 & \frac{1}{3} & -\frac{3}{2} \\ 0 & 0 & 1 & -\frac{1}{3} & 2 \end{array}\right]$$

R.E.F. $\rightsquigarrow$ $\begin{cases} x_1 - \frac{1}{2}x_2 + \frac{1}{3}x_4 = -\frac{3}{2} \\ x_3 - \frac{1}{3}x_4 = 2 \end{cases}$

* x_1, x_3 basic, x_2, x_4 free $\Rightarrow$

$$\begin{cases} x_1 = \frac{1}{2}x_2 - \frac{1}{3}x_4 - \frac{3}{2} \\ x_2 = \text{free} \\ x_3 = \frac{1}{3}x_4 + 2 \\ x_4 = \text{free} \end{cases}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_2 - \frac{1}{3}x_4 - \frac{3}{2} \\ x_2 \\ \frac{1}{3}x_4 + 2 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -\frac{1}{3} \\ 0 \\ \frac{1}{3} \\ 1 \end{bmatrix} + \begin{bmatrix} -\frac{3}{2} \\ 0 \\ 2 \\ 0 \end{bmatrix}$$

2 . [5 pts] Is the following linear system consistent or inconsistent? If it is consistent, find its general solution.

$$\begin{cases} 2x_1 + x_2 + 3x_4 + x_5 = 0 \\ 2x_1 + 2x_3 + 4x_4 + 4x_5 = 4 \\ x_1 + x_2 - 4x_3 + x_4 - x_5 = -2 \end{cases}$$

We find the REF of the augmented matrix:

$$\left[\begin{array}{ccccc|c} 2 & 1 & 0 & 3 & 1 & 0 \\ 2 & 0 & 2 & 4 & 4 & 4 \\ 1 & 1 & -4 & 1 & -1 & -2 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccccc|c} 1 & 1 & -4 & 1 & -1 & -2 \\ 2 & 0 & 2 & 4 & 4 & 4 \\ 2 & 1 & 0 & 3 & 1 & 0 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccccc|c} 1 & 1 & -4 & 1 & -1 & -2 \\ 0 & -2 & 10 & 2 & 6 & 8 \\ 0 & -1 & 8 & 1 & 3 & 4 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccccc|c} 1 & 1 & -4 & 1 & 1 & -2 \\ 0 & -1 & 8 & 1 & 3 & 4 \\ 0 & -2 & 10 & 2 & 6 & 8 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \left[\begin{array}{ccccc|c} 1 & 1 & -4 & 1 & -1 & -2 \\ 0 & -1 & 8 & 1 & 3 & 4 \\ 0 & 0 & -6 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_3 \to -\frac{1}{6} R_3} \left[\begin{array}{ccccc|c} 1 & 1 & -4 & 1 & -1 & -2 \\ 0 & -1 & 8 & 1 & 3 & 4 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow[R_2 \to R_2 - 8R_3]{R_1 \to R_1 + 4R_3} \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 1 & -1 & -2 \\ 0 & -1 & 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{array}\right] \xrightarrow[R_2 \to -R_2]{R_1 \to R_1 + R_2} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 2 & 2 & 2 \\ 0 & 1 & 0 & -1 & -3 & -4 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{array}\right]$$

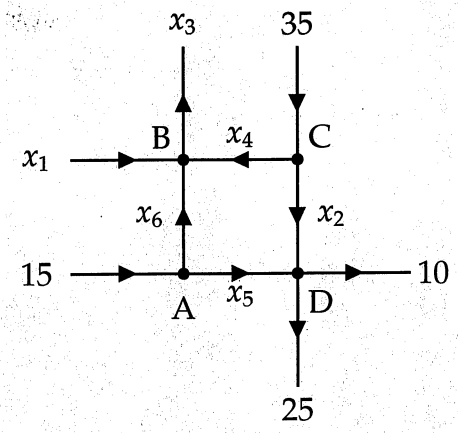
R.E.F.

* x_1, x_2, x_3 basic and x_4, x_5 free.

The system is consistent since the rightmost column is not a pivot column.

$$\begin{cases} x_1 + 2x_4 + 2x_5 = 2 \\ x_2 - x_4 - 3x_5 = -4 \\ x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -2x_4 - 2x_5 + 2 \\ x_2 = x_4 + 3x_5 - 4 \\ x_3 = 0 \\ x_4 = \text{free} \\ x_5 = \text{free} \end{cases}$$

3. **[3 pts]** Write down a system of equations describing the following traffic flow problem. The letters A through D label intersections and numerical values indicate flow in cars per minute. The arrows indicate the direction of flow (all roads are one-way). Include all relevant equations. You do **not** need to find the general flow pattern, i.e., you do **not** need to solve the linear system.



1) Total flow in = Total flow out

$$x_1 + 35 + 15 = x_3 + 10 + 25$$

2) At every intersection, flow in = flow out

(A) $15 = x_5 + x_6$

(B) $x_1 + x_4 + x_6 = x_3$

(C) $35 = x_2 + x_4$

(D) $x_2 + x_5 = 10 + 25$

4. **[4 pts]** Suppose a linear system has augmented matrix

$$\left[\begin{array}{cc|c} 1 & -2 & a \\ 3 & b+2 & -2 \end{array}\right]$$

for some real numbers a and b. For which values of a and b does the system have:

(a) No solution?
(b) Exactly one solution?
(c) Infinitely many solutions?

Your answers should include all possibilities for the values of a and b. Write your final answer in the spaces at the bottom of the page.

We row reduce the matrix:

$$\left[\begin{array}{cc|c} 1 & -2 & a \\ 3 & b+2 & -2 \end{array}\right] \xrightarrow{R_2 \to R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & -2 & a \\ 0 & b+8 & -2-3a \end{array}\right]$$

inconsistent: $b+8 = 0$ and $-2-3a \neq 0 \Rightarrow \begin{cases} a \neq -\frac{2}{3} \\ b = -8 \end{cases}$

unique solution: $b+8 \neq 0 \Rightarrow \begin{array}{l} b \neq -8 \\ a \text{ arbitrary} \end{array}$

infinitely many solutions: $b+8 = 0$ and $-2-3a = 0 \Rightarrow \begin{cases} b = -8 \\ a = -\frac{2}{3} \end{cases}$

Final Answer: (a) $\begin{cases} b = -8 \\ a \neq -\frac{2}{3} \end{cases}$ **(b)** $\begin{cases} b \neq -8 \\ a \text{ arbitrary} \end{cases}$ **(c)** $\begin{cases} b = -8 \\ a = -\frac{2}{3} \end{cases}$

5. **[4 pts]** Determine if the following vectors are linearly dependent or linearly independent. Justify your answer.

$$\begin{bmatrix}1\\-2\\3\end{bmatrix} \quad \begin{bmatrix}-2\\2\\-1\end{bmatrix} \quad \begin{bmatrix}-4\\2\\3\end{bmatrix}$$

We row reduce the matrix below:

$$\left[\begin{array}{ccc|c} 1 & -2 & -4 & 0 \\ -2 & 2 & 2 & 0 \\ 3 & -1 & 3 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 + 2R_1} \left[\begin{array}{ccc|c} 1 & -2 & -4 & 0 \\ 0 & -2 & -6 & 0 \\ 0 & 5 & 15 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + \frac{5}{2}R_2} \left[\begin{array}{ccc|c} 1 & -2 & -4 & 0 \\ 0 & -2 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_2 \to -\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & -2 & -4 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 + 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \text{R.E.F.}$$

The third column is not a pivot column. Therefore the homogeneous linear system corresponding to the original augmented matrix has nontrivial solutions, i.e., the vectors are <u>linearly dependent</u>

6. **[4 pts]** Consider the following vectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ -6 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

Is $\mathbf{w}$ in Span$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$? Justify your answer.

We row reduce the matrix $[v_1 \; v_2 \; v_3 \mid w]$.

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ -2 & 2 & 1 & -1 \\ 3 & -6 & -4 & 3 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 + 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & 6 & 5 & 1 \\ 0 & -12 & -10 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & 6 & 5 & 1 \\ 0 & 0 & 0 & 2 \end{array}\right] \quad \text{E.F.}$$

Since the rightmost column is a pivot column, w is <u>not</u> in Span $\{v_1, v_2, v_3\}$.